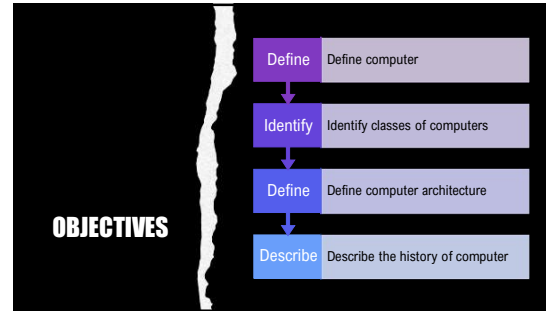
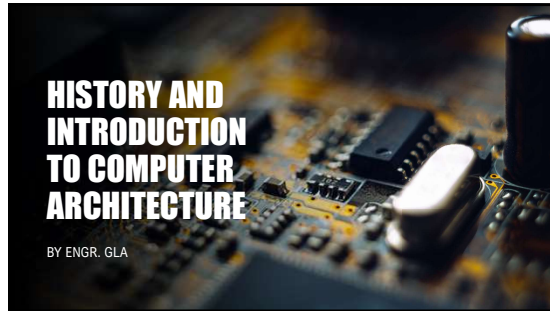




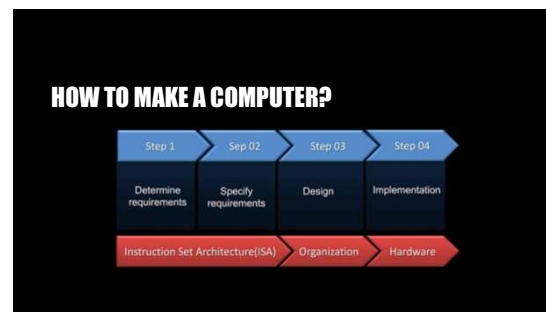
CLASSIFICATIONS OF COMPUTERS

A. Essential of the material

1. Mechanics – chassis, cooling systems, hard drive spindles
2. Electronics – CPU, GPU, memory, and other integrated circuits that process data and execution instruction
3. Biology – DNA computing, bioinformatics

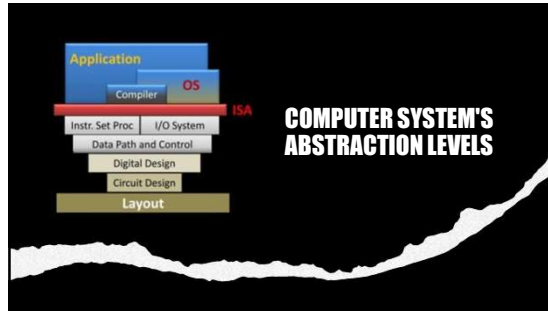
B. How it is used

1. Desktop Computer – a personal computer designed for regular use
2. Server Computer – a computer designed to process requests and deliver data to other computers over a network, serving multiple users and managing large-scale operations.
3. Embedded Computer - a specialized computer that is integrated into other devices, performing specific control functions within a larger system



WHAT IS COMPUTER ARCHITECTURE?

- Programmer's view: ISA – Amdahl, 1964
- Design of Computer system from its instruction set and structure down to function units – Harold S. Stone, 1987
- System concept integrating hardware, software algorithm and languages – Hwang and Briggs, 1984
- Instruction Set Design, Organization, and Hardware – Hennessy and Patterson, 1990



COMPUTER SYSTEM'S ABSTRACTION LEVELS

- **Layout** : physical arrangement of circuits and components on a chip
- **Circuit Design**: involves creating the circuits that perform the basic operations of a computer, like adding numbers or storing bits
- **Digital Design**: this level focuses on designing digital components like logic gates that execute instructions
- **Data Path and Control**: design the flow of data and the control signals that guide the operations within the processor

COMPUTER SYSTEM'S ABSTRACTION LEVELS

- **ISA**: defines the basic commands that a processor can execute
- **I/O System**: deals with how the computer communicates with external devices like keyboards or printers.
- **Compiler**: translates high-level programming languages into machine code
- **OS**: manages the hardware and software resources of the computer and provides services for programs
- **Application**: programs or software that users can interact with directly, like web browsers, games, or word processors.

HISTORY OF COMPUTERS

0th Generation (- 1900) : MECHANICS



Abacus

- Abacus considered first mechanical computing device
- Used beads and rods to count numbers.

HISTORY OF COMPUTERS

0th Generation (- 1900) : MECHANICS

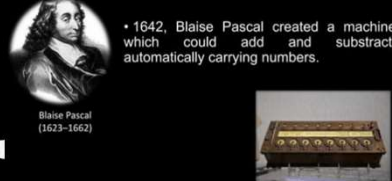


William Oughtred (1575-1660)

- 1612, John Napier used floating point arithmetic and invented the logarithm
- 1622, William Oughtred created the slide rule based on Napier's logarithms.

HISTORY OF COMPUTERS

0th Generation (- 1900) : MECHANICS



Blaise Pascal (1623-1662)

- 1642, Blaise Pascal created a machine which could add and subtract, automatically carrying numbers.

HISTORY OF COMPUTERS

0th Generation (- 1900) : MECHANICS



Gottfried Leibniz
(1646-1716)

- 1673, Gottfried Leibniz built a calculator which could multiply as well.



HISTORY OF COMPUTERS

0th Generation (- 1900) : MECHANICS



Joseph-Marie Jacquard
(1752-1834)

- He invented an automatic loom using punched cards to control patterns in the fabrics.
- Starting the industrial age



HISTORY OF COMPUTERS

0th Generation (- 1900) : MECHANICS



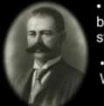
Charles Babbage
(1752-1834)

- 1822, He designed the Difference Engine which can compute navigational tables.
- 1833, He designed the Analytical Engine
- 1991, it is built by the Science Museum in London



HISTORY OF COMPUTERS

0th Generation (- 1900) : MECHANICS



Herman Hollerith
(1860-1929)

- He developed a mechanical tabulator based on punched cards to rapidly tabulate statistics from millions of pieces of data
- Founded Hollerith Tabulating Company Which became IBM in 1924



HISTORY OF COMPUTERS

1st Generation (1900 - 1957) : VACUUM TUBES



Konrad Zuse
(1910-1995)

- 1935 - 38, He developed Z1 and Z2 mechanical computers using binary arithmetic.



HISTORY OF COMPUTERS

1st Generation (1900 - 1957) : VACUUM TUBES



John Vincent Atanasoff
(1903-1995)

- 1936 - 39, They built Atanasoff-Berry Computer(ABC) for solving linear systems in physics. Introduced ALU and rewriting memory



Clifford Berry
(1918-1963)
Introduction to Computer Architecture

HISTORY OF COMPUTERS

1st Generation (1900 - 1957) : VACUUM TUBES



Alan Turing
(1912-1954)

- 1943, Alan Turing built Colossus which is used to break German codes (is encrypted by ENIGMA machine).



HISTORY OF COMPUTERS

1st Generation (1900 - 1957) : VACUUM TUBES



Howard Hathaway Aiken
(1900-1973)

- 1944, Harvard Mark 1 is built, which is used to compute artillery and navigation tables



HISTORY OF COMPUTERS

1st Generation (1900 - 1957) : VACUUM TUBES

- ENIAC – The first Electronic Computer
- 1943, work started on ENIAC at University of Pennsylvania under John Mauchly and J. Presper Eckert with Herman Goldstein



HISTORY OF COMPUTERS

1st Generation (1900 - 1957) : VACUUM TUBES

Electronic Numerical Integrator and Computer



- Used 18000 vacuum tubes
- U shaped, 25m long, 2.5m high, 1.5m wide
- Programmed by plugging cables and setting switches
- From 1 hour to 1 day to program

HISTORY OF COMPUTERS

1st Generation (1900 - 1957) : VACUUM TUBES



John von Neumann
(1900-1973)

- 1944, He joined to ENIAC team
- Credited with the idea of storing programs as numbers
- 1945, He proposed a stored program computer called EDVAC

HISTORY OF COMPUTERS

2nd Generation (1957 - 1965) : TRANSISTORS

- 1958, Integrated Circuit invented
- 1960, DEC PDP-1, Precursor to the Minicomputer
- 1965, IBM/360 Mainframe is introduced using integrated circuits
- 1965, DEC introduced PDP-8, first minicomputer



HISTORY OF COMPUTERS

3rd Generation (1966 - 1980) : INTEGRATED CIRCUITS

- 1976, Steve Jobs and Steve Wozniak develop Apple I in their parent's garage.
- 1976, Cray-1: The first supercomputer announced
- 1977, TRS-80 released by Tandy Corporation



HISTORY OF COMPUTERS

3rd Generation (1966 - 1980) : INTEGRATED CIRCUITS

- Integrated Circuit(IC): Small Scale Integration(SSi), Medium Scale Integration(MSI), Large Scale Integration(LSI).
- 1971, Intel 4004 the first microprocessor
- 1973, Xerox invents Ethernet
- 1975, First PC, MITS Altair 8800, Bill Gates and Paul Allen wrote a BASIC compiler for the Altair.



HISTORY OF COMPUTERS

4th Generation (1981 - PRESENT) : MICROPROCESSORS

- Very Large Scale Integration (VLSI) and Super Large Scale Integration(SLSI)
- IBM PC 1981, Intel 8088 and DOS OS
- 1981, First "Portable Computer"



HISTORY OF COMPUTERS

4th Generation (1981 - PRESENT) : MICROPROCESSORS

- 1982, Commodore 64 released
- Sold in Retail Stores
- 10 000 commercial software titles
- Best-selling single personal computer model of all time



HISTORY OF COMPUTERS

4th Generation (1981 - PRESENT) : MICROPROCESSORS

- 1991 – First 64bit Microprocessor produced by MIPS Technologies – R4000
- 2000 – IBM Releases first "Dual-Core" CPU –Power4
- 2003 – 64 bits brought mainstream with AMD64
- 2004 – Intel follows with EMT64
- 2005 – Intel releases Pentium D Dual-Core 64 bit processor

HISTORY OF COMPUTERS

5th Generation
(PRESENT - FUTURE) :
ARTIFICIAL INTELLIGENCE

